BUS105

Examination - January Semester 2024

Statistics

Thursday, 18 April 2024 **10:00 am - 12:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **4** questions and comprises **8** printed pages (including cover page).
2. You must answer **4** questions.
3. All answers must be written in the answer book.
4. This is **Open-book** examination.

At the end of the examination.
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.
If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

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(Full marks: 100)

Question 1

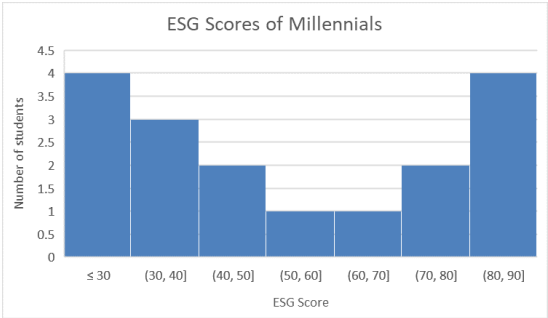
Singapore Government and companies have been actively addressing environmental, social and governance (ESG) issues in the last three years. Studies showed that companies that emphasize ESG issues are more likely to engage and motivate employees. Nevertheless, there has been a general perception that younger workers may be the ones that place more value in ESG initiatives.

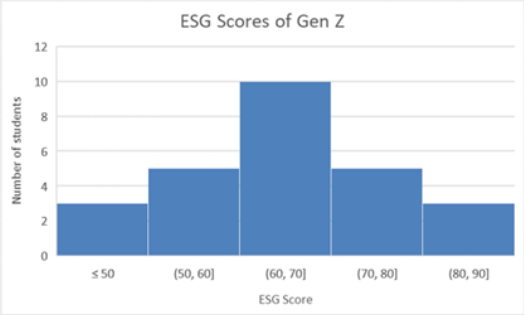
Professor Tan from SUSS is teaching Sociology and he wants to determine whether this is indeed true. He has a class of part-time students this year and decided to divide them into two groups. The first group is known as Gen Z and consists of students who were born after 1997 (less than 26 years old). The second group is known as Millennials and consists of students born between 1981 to 1996 (27 to 42 years old). There were a few students who were born before 1981 and Prof Tan has included them in the Millennials for the ease of grouping.

Prof Tan developed a scale to measure students' valuation of ESG. The scale is from 0 to 100 with students having high scores means the ESG initiatives of companies are important to them. The following table and charts provide a summary of the results:

ESG Score		
	Gen Z	Millennials
Mean	66.2	54.1
Standard Error	2.0	5.8
Median	66	45
Mode	58	34
Standard Deviation	10.4	23.8
Sample Variance	108.7	566.6
Kurtosis	-0.33	-1.85
Skewness	0.14	0.18
Range	37	63
Minimum	48	25
Maximum	85	88
Sum	1722	919
Count	26	17

Table 1.1: Summary statistics of ESG scores from Gen Z and Millennial students





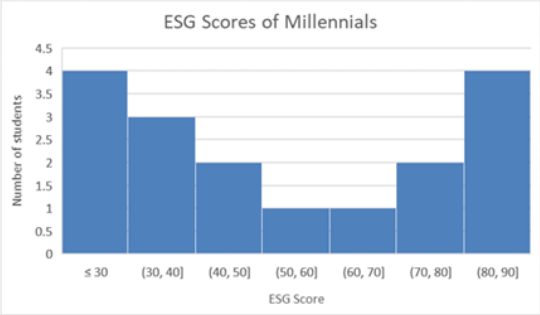


Figure 1.1: Histograms comparing the ESG scores of Gen Z and Millennial students

Question 1a

Interpret the results presented and summarise these as an executive summary. You should assist Prof Tan to draw a preliminary conclusion base on your observation. Please keep your summary to **within 300 words**.

(15 marks)

Question 1b

Explain why these results are **NOT** statistically acceptable to support Prof Tan's research. You are required to provide ***three (3)*** statistical reasons to justify your view.

(10 marks)

Question 2

Question 2a

The following is the probability distribution of a random variable X:

$X = x$	1	3	4	5
$P(X = x)$	0.28	0.11	k	0.13

i. Identify the value of k.

ii. Determine the probability of $X + \sqrt{X}$ is less than 5.

iii. Find the standard deviation of X.

(12 marks)

Question 2b

The monthly household income for people staying in 3-room flats in Singapore can be assumed to be normally distributed with mean \$5,700 and standard deviation \$1,600.

i. If a household staying in a 3-room flat in Singapore is randomly selected, identify the probability that its monthly household income is between \$5,000 and \$7,000.

ii. If 30 households staying in 3-room flats in Singapore are randomly selected, report the probability that their mean monthly household income is more than \$6,000.

(13 marks)

Question 3

Delight Perfume Pte Ltd launched a new advertisement recently. The Marketing Manager wants to assess the effectiveness of the new advert. In particular, he suspects that the advert is more effective on women than men. He randomly selected 30 male and 30 female customers to participate in a survey. They were asked to complete a questionnaire that measures their awareness of the advert. The total score is 100 and customers who can recall a lot of details will obtain a high score. The summary statistics for the data are given in ***Table 3.1***, and the test result is given in ***Table 3.2***.

Awareness Score		
	Male	**Female**
Mean	69.5	77.7
Standard Error	2.2	3.2
Median	74.5	83.5
Mode	81	91
Standard Deviation	12.0	17.5
Sample Variance	144.6	306.9
Kurtosis	-0.850	-0.636
Skewness	-0.706	-0.760
Range	37	61
Minimum	45	38
Maximum	82	99
Sum	2085	2331
Count	30	30
Confidence Level(95.0%)	4.5	6.5

Table 3.1: Summary statistics of customers' awareness scores

	Male	*Female*
Mean	69.5	77.7
Variance	144.6	306.9
Observations	30	30
Pooled Variance	225.8	
Hypothesized Mean Difference	0	
df	58	
t Stat	-2.11369	
P(T<=t) one-tail	0.019427	
t Critical one-tail	1.671553	
P(T<=t) two-tail	0.038854	
t Critical two-tail	2.001717	

Table 3.2: Result of an appropriate hypothesis test

Question 3a

Describe the 95% confidence intervals for the mean awareness scores of male and female customers, respectively. Based on the confidence intervals, discuss whether we can conclude that the mean awareness scores for female customers is higher than male customers.

(8 marks)

Question 3b

Apply an appropriate hypothesis test at the 5% significance level to determine whether the mean awareness scores for female customers is higher than male customers.

(10 marks)

Question 3c

Compare the results from Part ***(a)*** and Part ***(b)***. Should the conclusion be different or similar? Please provide your justification.

(7 marks)

Question 4

Zhongqing Group runs a chain of hotels in various cities in China. The Group wishes to use the data from its existing hotels to determine a good location for a new hotel. The Planning Director randomly selected 40 hotels from its group and collected the following information:

Sales: Annual sales in millions of RMB

Competitors: Number of competitors in the city where the hotel is located

Rooms: Number of rooms available in the hotel

Pool: Availability of indoor swimming pool: 0 means 'No', 1 means 'Yes'

A multilinear regression analysis was conducted and the following is the statistical output:

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.632962895
R Square	0.400642026
Adjusted R Square	0.353324291
Standard Error	7.348563625
Observations	42

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	3	1371.698709	457.2329	8.467058	0.000196386
Residual	38	2052.052719	54.00139		
Total	41	3423.751429			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	-65.6263446	29.8617981	-2.19767	0.034139	-126.0783944	-5.174294814
Competitors	-0.04110056	0.466870438	-0.08803	0.930312	-0.986230348	0.904029232
Rooms	0.052810694	0.01270846	4.155554	0.000178	0.027083761	0.078537626
Pool	4.341293892	2.355632748	1.842942	0.073149	-0.427435295	9.11002308

Table 4.1: Statistical report for multiple linear regression analysis

Question 4a

Express the relationship between sales and the ***three (3)*** independent variables by writing down the linear equation. You may keep the value of the coefficients to ***three (3)*** significant figures. Interpret the coefficients to show how each of the independent variables impacts sales.

(5 marks)

Question 4b

The Planning Director is examining a location which can build a hotel for 2,000 rooms. He intended to have an indoor swimming pool for this hotel. He discovered that there are currently eight competitors operating in that city. Estimate the sales of this new hotel if it is built.

(4 marks)

Question 4c

Execute appropriate hypothesis tests to advise the Planning Director on which variable(s) is/are statistically significant at the 5% significance level. Write down your steps in detail.

(10 marks)

Question 4d

Based on the results, offer ***two (2)*** recommendations to the Planning Director to improve the regression analysis.

(6 marks)

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